

Note: As crm.hellogrowthcrm.com was inaccessible from 20th-24th Sept 2026 due to ERR_ADDRESS_UNREACHABLE error, this document is prepared based on standard CRM architecture and Shopify public documentation. Screenshot attached as proof.

1. Objective

Build real-time 2-way sync between Shopify store and HelloGrowthCRM to automate lead nurturing, sales pipeline, and customer retention. Reduce manual data entry by 90%.

2. Shopify APIs Required

- Customers API: GET /admin/api/2024-01/customers.json - For contact creation
- Orders API: GET /admin/api/2024-01/orders.json - For deal pipeline
- Products API: GET /admin/api/2024-01/products.json - For catalog sync
- Checkouts API: GET /admin/api/2024-01/checkouts.json - For abandoned cart leads

3. Authentication Flow

Use OAuth 2.0. Merchant clicks "Connect Shopify" in CRM > Redirects to Shopify > Approves scopes: read_customers, read_orders, read_checkouts > Access token stored encrypted in MongoDB.

4. Webhooks for Real-time Sync

- customers/create → Create/Update CRM Contact + Trigger Welcome Email
- orders/create → Create Deal in "New Order" pipeline stage
- orders/paid → Move Deal to "Won" + Send Invoice
- checkouts/update → If abandoned > Create Lead + Schedule 1hr reminder email

5. Tech Stack: MERN

Backend: Node.js + Express microservice to handle webhooks and API calls.

Use `shopify-api-node` package. Queue system with BullMQ for rate limit handling.

Database: MongoDB to store tokens, sync logs, and mapping. Unique index on email.

Frontend: React component in CRM Settings > Integrations for Shopify connect, field mapping, and error logs UI.

6. Error & Duplicate Handling

Rate Limit 429: Exponential backoff retry 1s, 2s, 4s. After 3 fails, log in CRM.

Duplicate: Match by email+phone. If exists, update. Else create new.

Store shopify_customer_id to prevent double sync.

7. Security & Privacy

All tokens AES-256 encrypted. Webhooks verified via HMAC-SHA256.

GDPR Delete: API call to remove customer from Shopify + CRM both.

8. Business Impact

- Lead Nurturing: Abandoned cart auto-email recovers 15% sales
- Segmentation: Tag customers by total_spent > 10k = VIP for campaigns

- Sales Pipeline: Orders > 5k auto-marked "Hot Deal" for team alerts
- Retention: Refund event tags customer "Churn Risk" for support call

9. Challenges & Solutions

Challenge: Shopify 2 calls/sec rate limit. Solution: Bulk Operations API + Queue.

Challenge: Historical data import. Solution: One-time script for past 6 months orders.

Challenge: Field mismatch. Solution: UI for admin to map Shopify fields to CRM fields.

10. Additional Suggestion

Integrate WhatsApp API via Interakt. On 'order/shipped' webhook, send auto WhatsApp update to customer. Reduces "where is my order" tickets by 60%.

11. 4-Week Roadmap

Week 1: OAuth + Customer/Order one-way sync

Week 2: Webhooks + Abandoned Cart automation

Week 3: React Settings UI + Error handling + Logs

Week 4: Product/Refund sync + Testing + Deployment

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